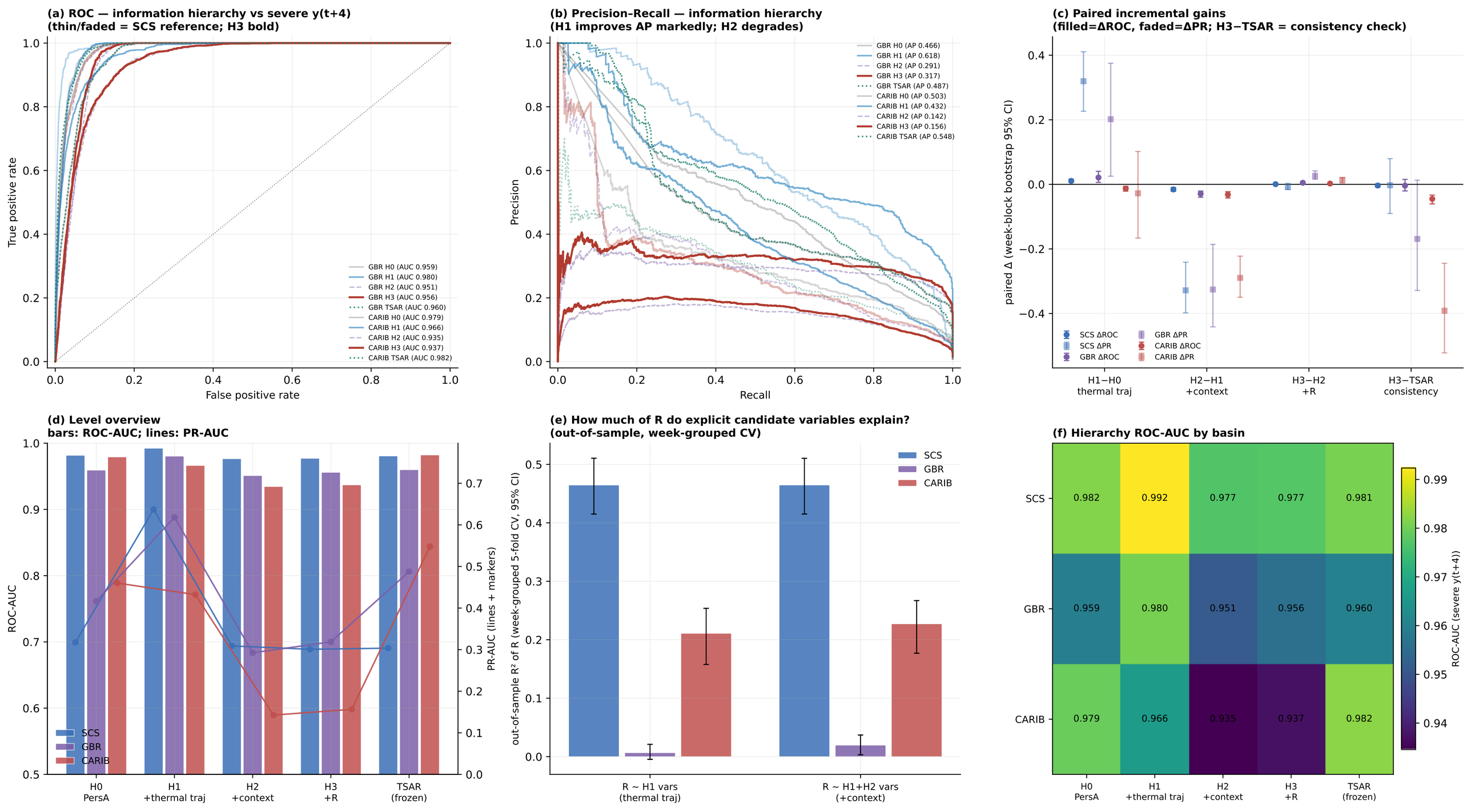

(a) ROC — information hierarchy vs severe y(t+4)
(thin/faded = SCS reference; H3 bold)
True positive rate
False positive rate
GBR H0 (AUC 0.959)
GBR H1 (AUC 0.980)
GBR H2 (AUC 0.951)
GBR H3 (AUC 0.956)
GBR TSAR (AUC 0.960)
CARIB H0 (AUC 0.979)
CARIB H1 (AUC 0.966)
CARIB H2 (AUC 0.935)
CARIB H3 (AUC 0.937)
CARIB TSAR (AUC 0.982)
(b) Precision–Recall — information hierarchy
(H1 improves AP markedly; H2 degrades)
Precision
Recall
GBR H0 (AP 0.466)
GBR H1 (AP 0.618)
GBR H2 (AP 0.291)
GBR H3 (AP 0.317)
GBR TSAR (AP 0.487)
CARIB H0 (AP 0.503)
CARIB H1 (AP 0.432)
CARIB H2 (AP 0.142)
CARIB H3 (AP 0.156)
CARIB TSAR (AP 0.548)
(c) Paired incremental gains
(filled=ΔROC, faded=ΔPR; H3−TSAR = consistency check)
paired Δ (week-block bootstrap 95% CI)
SCS ΔROC
SCS ΔPR
GBR ΔROC
GBR ΔPR
CARIB ΔROC
CARIB ΔPR
H1−H0 thermal traj
H2−H1 +context
H3−H2 +R
H3−TSAR consistency
(d) Level overview
bars: ROC-AUC; lines: PR-AUC
ROC-AUC
PR-AUC (lines + markers)
SCS
GBR
CARIB
H0 PersA
H1 +thermal traj
H2 +context
H3 +R
TSAR (frozen)
(e) How much of R do explicit candidate variables explain?
(out-of-sample, week-grouped CV)
out-of-sample R² of R (week-grouped 5-fold CV, 95% CI)
SCS
GBR
CARIB
R ~ H1 vars (thermal traj)
R ~ H1+H2 vars (+context)
(f) Hierarchy ROC-AUC by basin
SCS
GBR
CARIB
0.982
0.992
0.977
0.977
0.981
0.959
0.980
0.951
0.956
0.960
0.979
0.966
0.935
0.937
0.982
H0 PersA
H1 +thermal traj
H2 +context
H3 +R
TSAR (frozen)
ROC-AUC (severe y(t+4))